

Adam Torres Encarnacion

University Park, PA | [Email](#) ✉ | [LinkedIn](#)  | [GitHub](#)  | [Portfolio](#) ▲

EDUCATION

The Pennsylvania State University

B.S. in Data Science, B.S. in Statistics

Expected: Dec 2026

GPA: 3.7

WORK EXPERIENCE

Amazon

May 2026 – Aug 2026

Software Engineering Intern

Cupertino, CA

- Refactored C++ infrastructure for a dual-arm fiber-cleaning robot targeting \$28.6M/year in savings across 85 data centers
- Built a 19-route HTTP server for remote robot control over SSH, maintaining a persistent sub-2ms on-device control loop
- Implemented a VLM-as-judge pipeline for live detection of 6 optical fibers & transceivers, centralizing detection data in S3
- Bootstrapped an agent-first, self-evolving FastMCP knowledge base & migrated legacy docs for 2 repos (~400 files)
- Hardened CI/CD via 2 internal AWS pipeline stages, blocking lints, complexity checks, & ~40 integration tests

IBM

May 2025 – May 2026

AI Engineering Intern

Remote

- Shipped a REST LLM microservice that generated documentation for 3100+ scripts and reduced team costs by \$223,000
- Helped triage ~500k criminal records spanning 20+ years with a LangChain chatbot built on Python, cutting review time
- Deployed serverless Azure Function Apps with automatic codebase scaling, cutting the manual review workload by 78%
- Refined an AI file-chunking pipeline with Llama-4 and GPT-4o-mini to process 7+ large-scale internal requests

PROJECTS

 **YADL (OpenAI WebMCP Challenge '26)** | *Python, OpenCV, MPS, PostgreSQL, AWS, Vercel, Railway*

- Turned AI agents into click-free CV labeling software through 26 WebMCP tools to draw & navigate an open canvas
- Emulated Codex Micro via OpenCV, hitting 98.78% F1 on 1,253 AI-labeled images after 600 MPS-accelerated epochs
- Balanced CV datasets with 8 deterministic image augmentations and GPT-Image-2 in async augmentation jobs
- Supported 16 concurrent labeling sessions at <300 ms p95 metadata API latency using FastAPI, PostgreSQL, and S3

 **nexdraw (NexHacks '26 ByteDance 2nd place winner)** | *TypeScript, Next.js, Gemini, PostgreSQL*

- Placed 2nd of 1500+ solo with a tldraw canvas that auto-completes sketches in place via Gemini AI image generation
- Classified chat vs draw with Gemini, then cropped the subject with Hugging Face Grounding DINO on CUDA (PyTorch)
- Piped voice prompts through OpenAI Whisper and gated every generation behind an accept/reject overlay on the canvas
- Persisted tldraw snapshots as PostgreSQL JSONB on a 2s debounce, writing PNG previews and skipping saves offline

 **simulacra (YHack '26 K2 Think V2 1st place winner)** | *Python, FastAPI, LangGraph, React*

- Designed a multi-agent orchestration engine in FastAPI with LangGraph StateGraph, running concurrent cognitive loops via asyncio with shared memory streams and Pydantic-validated structured outputs
- Built per-agent memory retrieval with temporal decay scoring, threshold-triggered reflection, and Deffuant bounded-confidence opinion dynamics to model echo-chamber formation across 50+ agents streamed over Socket.IO

LEADERSHIP EXPERIENCE

Claude Builder Club

Aug 2025 – Jan 2026

Vice President

University Park, PA

- Brought Claude AI to Penn State, drawing in 260+ club signups in its first month (the fastest growth in school history)
- Taught AI tooling (agent harnesses, skills, hooks, MCPs, multiplexers) to 170+ students, with 90% returning for follow-ups

Nittany AI

Jan 2025 – Jan 2026

Director (AI)

University Park, PA

- Led a research reading group for the largest AI club on campus, with 40+ students per semester and 85% retention
- Covered papers [YOLO](#), [SAM3](#), [Attention Is All You Need](#), and [Adam Optimizer](#) across 5 months of student-led seminars

TECHNICAL SKILLS

Programming Languages: Python, C++, C, C#, SQL, TypeScript, HTML/CSS, R, Lua

Frameworks: PyTorch, OpenCV, Hugging Face, Pandas, NumPy, FastAPI, React, Node.js, Next.js, LangChain/LangGraph

Developer Tools: Git, Linux, VS Code, Vim/Neovim, Pi (harness), Cmux (terminal multiplexer), Agile/Scrum

Databases: PostgreSQL, MySQL, SQLite, Redis, MongoDB, DynamoDB

Cloud/Infra: AWS (Lambda, S3, EC2, Bedrock, DynamoDB), Azure Functions, Vercel, Docker, CI/CD (GitHub Actions)